

Chapter 1

Right-of-Way Costs Calculation Documentation

1.1 Introduction

The `row_costs.py` script calculates right-of-way (ROW) costs for a transmission line project. ROW costs depend on the **ROW agreement type**: who owns or controls the corridor and how the utility secures the right to use it. The script uses a **ROW agreement type** switch to determine which cost terms apply, so that acquisition (upfront payment) and annual ROW payment (ongoing permission) are **mutually exclusive** unless the regime explicitly allows both (e.g., federal/hybrid).

Cost terms used in the model:

- **Acquisition**: One-time upfront payment (easement compensation, fee-simple purchase, or similar).
- **Holding cost**: Option fee—an explicit annual payment during the delay (permitting) period to secure the ROW so it is not taken by others. Not the same as “cost of capital” on acquisition; cost of capital on acquisition during delay is handled by AFUDC.
- **Annual ROW payment**: Ongoing payment for permission to use the corridor (lease, license, or federal annual use fee). Labeled in data and outputs as “annual ROW payment”; in legacy inputs the parameter may still be named `rent_cost`.

Canonical notation. The canonical variable names and equations for ROW are in `COST METHODOLOGY.md`: Capital ROW (acquisition and holding) under “Capital ROW costs (1.b)”, and operational ROW rent under “Operational ROW rent (2.c)”. Capital uses $C_{\text{ROW,capital,nominal}}$, $C_{\text{ROW,capital,real}}$; rent uses $C_{\text{rent,annual}}$, T_{rent} , $t_{\text{rent,start}}$, $C_{\text{rent,total,nominal}}$, $C_{\text{rent,total,real}}$.

1.2 ROW Agreement Types

The following agreement types drive which cost terms are applied. Only one regime applies per scenario.

1.2.1 Permanent easement (new)

Typical greenfield: utility secures a **permanent** right to build and operate (easement or similar). Landowner or agency is paid upfront; there is no ongoing “rent” for the same corridor.

- **Acquisition:** Yes (upfront payment).
- **Annual ROW payment:** No (set to zero in the calculator).
- **Holding cost:** Optional. If the user provides an option fee (**hold_cost** per acre per year), it is applied during the delay period only.

1.2.2 Lease / license (existing)

Utility uses an **existing** corridor (e.g., highway, railroad, another utility) and pays the owner for ongoing permission. No upfront purchase of the ROW.

- **Acquisition:** No (set to zero).
- **Holding cost:** No (set to zero).
- **Annual ROW payment:** Yes (paid over delay + construction + project lifetime, or as configured).

Reconductoring or “uses existing ROW” scenarios typically use this regime.

1.2.3 Fee simple purchase

Utility **buys** the strip of land. Same cost treatment as permanent easement (new): upfront only, no annual ROW payment. Holding (option fee) optional during delay if applicable.

1.2.4 Federal / hybrid

Some federal or negotiated arrangements have both an upfront fee and an annual use fee for the same corridor. Both acquisition and annual ROW payment are applied.

- **Acquisition:** Yes.
- **Annual ROW payment:** Yes.
- **Holding cost:** Optional (option fee during delay if provided).

1.3 Mutual Exclusivity and Defaults

- For **permanent easement (new)** and **fee simple**, annual ROW payment from the ROW-details file is **ignored** (set to zero) so that “buy the ROW” does not also charge rent for the same miles.
- For **lease / license (existing)**, acquisition and holding are set to zero; only annual ROW payment applies.
- If `row_agreement_type` is not set in the project technical details, it is **inferred**: if `reconductoring` or `uses_existing_row` is true, regime is **lease_license_existing**; otherwise **permanent_easement_new**. This keeps existing scenarios consistent.

1.4 Configuring YAMLS for ROW Costs

This section describes how to set up the YAML files for each ROW scenario. No method re-explanation—only which files and keys to use.

Files involved:

- `01_project_technical_details.yaml` — project section: `reconductoring`, `uses_existing_row`, and optionally `row_agreement_type`.
- `11_project_row_details.yaml` — `right_of_way` with one entry per zone; each zone has `miles`, `acquisition_cost`, `hold_cost` (\$/acre/year), and `rent_cost` (\$/acre/year, annual ROW payment).

1.4.1 New greenfield (permanent easement)

Result: Acquisition + optional option fee (holding). No annual ROW payment for the same miles.

- **01:** `reconductoring: False`, `uses_existing_row: False`. Omit `row_agreement_type` (inferred as `permanent_easement_new`).
- **11:** Set `miles`, `acquisition_cost`, and `hold_cost` per zone. `rent_cost` is ignored. To get **acquisition only** (no option fee), set `hold_cost: 0` for every zone that has `miles > 0`.

1.4.2 Existing corridor (lease / license)

Result: Annual ROW payment only. No acquisition, no holding.

- **01:** `uses_existing_row: True` or `reconductoring: True`. Omit `row_agreement_type` (inferred as `lease_license_existing`).
- **11:** Set `miles` and `rent_cost` per zone. `acquisition_cost` and `hold_cost` are ignored.

1.4.3 Fee simple purchase

Result: Same as new greenfield (acquisition + optional holding; no annual ROW payment).

- **01:** `row_agreement_type: fee_simple`. Or use the same 01 as new greenfield (inference gives `permanent_easement_new`, which has the same cost treatment).
- **11:** Same as new greenfield: `acquisition_cost`, `hold_cost`; `rent_cost` ignored.

1.4.4 Federal / hybrid (upfront + annual)

Result: Acquisition and annual ROW payment (and optional holding). Both upfront and ongoing payments apply.

- **01:** `row_agreement_type: federal_hybrid`.
- **11:** Set miles, `acquisition_cost`, `rent_cost`, and optionally `hold_cost` per zone. All three cost fields are used.

1.4.5 Explicit override of inferred type

To force a regime that does not match the default inference (e.g., new build but you want lease treatment, or vice versa), set `row_agreement_type` in `01_project_technical_details.yaml` to one of: `permanent_easement_new`, `lease_license_existing`, `fee_simple`, `federal_hybrid`. Then configure `11_project_row_details.yaml` as in the corresponding subsection above.

1.4.6 Quick reference

Scenario	01 settings	11: which costs used
New greenfield	<code>reconductoring: False</code> , <code>uses_existing_row: False</code>	<code>acquisition_cost</code> , <code>hold_cost</code>
Acquisition only	Same as above	Same; set <code>hold_cost: 0</code> everywhere
Existing corridor	<code>uses_existing_row: True</code> or <code>reconductoring: True</code>	<code>rent_cost</code> only
Federal/hybrid	<code>row_agreement_type: federal_hybrid</code>	<code>acquisition_cost</code> , <code>rent_cost</code> , <code>hold_cost</code>

1.5 Method Overview

1.5.1 Purpose

The script:

1. Determines ROW width from project category.
2. Computes zone acres from zone miles and ROW width.
3. Loads ROW agreement type (or infers it).

4. For each zone with miles > 0, computes potential acquisition, holding, and annual ROW payment from per-acre rates; then applies the **agreement type** to zero out terms that do not apply.
5. Aggregates costs across zones.
6. Applies AFUDC capitalization (acquisition only, when eligible) for the regulatory perspective.
7. Computes present values for the societal perspective.

1.5.2 Acres and Zone-Level Potential Costs

Zone area (acres):

$$\text{Zone Acres} = \frac{\text{Zone Miles} \times 5280 \times \text{ROW Width (ft)}}{43560}. \quad (1.1)$$

From the ROW details file, each zone has (per acre): **acquisition_cost**, **hold_cost** (\$/acre/year), and **rent_cost** (\$/acre/year, interpreted as annual ROW payment). The script first computes:

$$\text{Potential acquisition} = \sum_{\text{zones}} \text{Zone Acres} \times \text{Acquisition cost/acre}, \quad (1.2)$$

$$\text{Potential yearly holding} = \sum_{\text{zones}} \text{Zone Acres} \times \text{Hold cost/acre/year}, \quad (1.3)$$

$$\text{Potential yearly annual ROW payment} = \sum_{\text{zones}} \text{Zone Acres} \times \text{Rent cost/acre/year}. \quad (1.4)$$

Then the **ROW agreement type** is applied:

- **Permanent easement (new) or fee simple**: Use acquisition and (optionally) holding; set annual ROW payment to zero.
- **Lease / license (existing)**: Use annual ROW payment only; set acquisition and holding to zero.
- **Federal / hybrid**: Use acquisition, annual ROW payment, and (optionally) holding.

Nominal totals (after applying the regime) are then computed for the chosen terms over the appropriate time periods (delay, construction, lifetime).

1.5.3 Inputs and Outputs

Inputs:

- Project technical details (construction type, AC/DC, capacity, etc.), including optional **row_agreement_type** (if absent, inferred from **reconductoring** and **uses_existing_row**).

- Physical details (terrain/zone miles).
- ROW width for project category (`20_project_category_row_widths`).
- ROW cost parameters by zone (`11_project_row_details`): `acquisition_cost`, `hold_cost`, `rent_cost` (annual ROW payment per acre per year).
- Financing (WACC, base year), AFUDC configuration, and cost timing patterns (`19_cost_timing_patterns`).
- Timeline (delay years, construction years, project lifetime).

Outputs:

- Nominal costs (acquisition, holding, annual ROW payment).
- AFUDC capitalized cost: ROW capital = acquisition + holding (when applicable); rent is excluded from capital.
- Present value costs (societal perspective); `row_capital_pv`, `row_rent_pv` for aggregation.
- Total acres.

Legacy output keys may still use “rent” (e.g., `rent_nominal`) for compatibility; conceptually the amount is “annual ROW payment.”

1.6 AFUDC, Capital vs. Operational

- **Acquisition** is AFUDC-eligible when the regime includes it; the cost timing pattern `row_acquisition` (e.g., 80% during delay, 20% during construction) is used to capitalize it to COD.
- **Holding cost** (option fee during delay) is **AFUDC-eligible**: it is capitalized using `row_holding` timing (100% during delay) so that ROW capital for rate base includes both acquisition and holding.
- **Annual ROW payment** (rent) is an **operational** expense: it is not AFUDC-eligible and is included in operational costs (with O&M and insurance), not in capital or rate base.
- **ROW capital** for rate base and for cost aggregation = acquisition + holding only. **Rate base** in CTCC = AFUDC capital (build + ROW capital + environmental mitigation).

1.7 Key Functions

1.7.1 calculate_zone_costs(row_width_feet)

Returns (yearly_holding_cost, acquisition_cost, yearly_rent_cost, total_acres) for all zones with miles > 0. Uses load_row_details() and miles_to_acres(miles, row_width_feet); sums zone acres × acquisition_cost, hold_cost, rent_cost per zone. The caller applies the ROW agreement type to zero out terms that do not apply.

Key logic:

```
1 row_details = load_row_details()
2 for zone_details in row_details["right_of_way"].values():
3     if zone_details["miles"] > 0:
4         zone_acres = miles_to_acres(zone_details["miles"],
5                                     row_width_feet)
6         acquisition_cost += zone_acres * zone_details["
7             acquisition_cost"]
8         yearly_rent_cost += zone_acres * zone_details["rent_cost"]
9         yearly_holding_cost += zone_acres * zone_details["hold_cost"]
10    ]
11 return yearly_holding_cost, acquisition_cost, yearly_rent_cost,
12        total_acres
```

1.7.2 main()

1. Load project technical details; construct category via build_category_string; get row_agreement_type from getattr(project_details, "row_agreement_type", None) or infer (lease_license_existing if reconductoring or uses_existing_row, else permanent_easement_new).
2. Load ROW width via load_row_widths(category); call calculate_zone_costs(row_width_feet).
3. Apply agreement type: lease_license_existing → acquisition = 0, holding = 0; permanent_easement_new or fee_simple → yearly_rent_cost = 0; federal_hybrid keeps all.
4. Compute nominal totals (holding × delay years; rent over appropriate years; acquisition_cost_used); rent_start_year and rent_total_years depend on agreement type (lease: year 1, delay+construction+lifetime; else: construction start, construction+lifetime).
5. Load load_afudc_setup(); if apply_afudc and acquisition used, calculate_afudc_capitalized_cost_timing_patterns["row_acquisition"], ...); else acquisition PV = acquisition / $(1 + r)^{\text{delay_years}}$.
6. add_row_costs(results) with total_nominal, total_afudc, total_pv, acquisition_nominal, holding_nominal, rent_nominal; write_batch_summary().

1.8 Present Value Timing

- **Holding cost:** Annually during delay (years 1 to `delay_years`).
- **Acquisition:** One-time at end of delay (year `delay_years`), discounted to base.
- **Annual ROW payment:** For lease/license (existing), from year 1 for (`delay` + `construction` + `lifetime`) years; for new builds with annual payment (federal/hybrid), from construction start for (`construction` + `lifetime`) years.

1.9 Acres Calculation

$$\text{Acres} = \frac{\text{Miles} \times 5280 \text{ ft/mile} \times \text{ROW Width (ft)}}{43560 \text{ ft}^2/\text{acre}}.$$

1.10 Integration

The script integrates with:

- `smart_loaders.py` — `load_project_technical_details`, `load_physical_details`, `load_row_width`, `load_row_details`, `load_financing_details`, `load_cost_timing_patterns`, `load_afudc_config`, `get_financing_data_raw`
- `calculation_utils.py` — `build_category_string`, `miles_to_acres`
- `financial_utils.py` — `calculate_present_value`, `calculate_afudc_capitalized_cost`, `validate_discount_rate`, `calculate_construction_start_year`, `load_afudc_setup`
- `path_config.py` — `YAMLS_DIR` (used by loaders)
- `smart_output.py` — `CTCCOutputManager` (`add_row_costs`, `write_batch_summary`)